Unit 1 :: The Lagrangian Approach

PHY0400104: Classical Mechanics

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Syllabus

Unit 1: Mechanics of point particles – the Lagrangian approach

Review of Newtonian mechanics; system of particles and constrained motion with types of constraints; degrees of freedom, generalised coordinates and velocities; principle of virtual work and D'Alembert's principle; Lagrange's (Euler-Lagrange, EL) equation; physical problems – simple and compound pendulums, simple harmonic oscillation, Lagrange's equations for a particle in spherical and cylindrical coordinate systems, falling body in uniform gravitational field, Lagrangian in central force problem.

1 Review of Newtonian mechanics

1.1 Kinematics

Kinematics is the branch of classical mechanics that describes the motion of points, objects, and systems of bodies without considering the forces that cause them to move. It focuses on the geometry of motion, analyzing parameters such as displacement, velocity, acceleration, and time to describe *how objects move*.

- **Particle**: A material body having mass but no spatial extent.
- **Position**: The position of a particle is usually specified by the vector $\boldsymbol{r}$ from the origin of a coordinate system to the particle.
- **Velocity**: The velocity of a particle is defined as the time rate of change of its position:

$$\boldsymbol{v} = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \dot{\boldsymbol{r}}$$

- **Acceleration**: The acceleration of a particle is defined as the time rate of change of its velocity:

$$\boldsymbol{a} = \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = \frac{\mathrm{d}^2\boldsymbol{r}}{\mathrm{d}t^2} = \ddot{\boldsymbol{r}}$$

- **Equation of motion**: It is a second-order differential equation for $\boldsymbol{a}$ as a function of $\boldsymbol{r}$, $\dot{\boldsymbol{r}}$ and t.
- **Trajectory**: It is the solution of the equation of motion which gives the position as a function of time, i.e. $\boldsymbol{r} = \boldsymbol{r}(t)$.

1.2 Dynamics

Dynamics is the branch of classical mechanics that describes how forces, torques, and energy cause, change, or maintain the motion of physical bodies. It explains *why objects move*, rather than just how they move. It is governed by Newton's Laws of Motion, Lagrangian mechanics, and Hamiltonian mechanics.

- **Newton's laws of motion**:
  1. *First law of motion*: a body not subjected to any external force will move in a straight line at constant speed.
  2. *Second law of motion*: the rate of change of momentum of a body is equal to the net external force applied to it.

$$\boldsymbol{F} \propto \frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t}$$

  3. *Third law of motion*: if a body exerts a force on a second body, the second body exerts and equal and opposite force of the same kind on the first one.

- **Conservation laws**: These laws are fundamental principles stating that certain physical properties remain constant over time within an isolated system.

  1. *Conservation of energy*: In an isolated system with no non-conservative forces (like friction), the total mechanical energy remains constant. Energy is transformed between kinetic and potential, and not created nor destroyed.
  2. *Conservation of linear momentum*: If the net external force on a system is zero, the total linear momentum ($\boldsymbol{p} = m\boldsymbol{v}$) remains constant.
  3. *Conservation of angular momentum*: If the net external torque on a system is zero, the total angular momentum ($\boldsymbol{L} = I\boldsymbol{\omega}$) remains constant.
  4. *Conservation of mass*: In non-relativistic dynamics (i.e. for speeds much smaller than the speed of light), the total mass of a closed system is conserved.

2 Constrained motion and virtual work

2.1 Generalized coordinates

In three-dimensional space we need three coordinates to specify the position of a single particle. For a system consisting of two particles, we need six coordinates.

Therefore, a system of N particles requires $3N$ coordinates. For N particles the positions of all the particles are

$$(x_1, y_1, z_1, x_2, y_2, z_2, \ldots, x_N, y_N, z_N)$$

.

To avoid being specific about the coordinate system we are using, we denote the coordinates by q_i and refer to them as *generalized coordinates*.

- If Cartesian coordinates are used then q_i take the values of (x, y, z).
- If spherical coordinates are used then q_i take the values of (r, θ, ϕ).
- If cylindrical coordinates are used then q_i take the values of (ρ, ϕ, z)

Notes:

1. It is often convenient to denote all the Cartesian coordinates by the letter x. Thus, for a single particle, (x, y, z) is written (x_1, x_2, x_3) and for N particles, $(x_1, y_1, \ldots, z_N)$ is expressed as $(x_1, x_2, \ldots, x_n)$, where $n = 3N$.
2. The dimensions of the generalized coordinates need not be those of length, and may even be dimensionless.

The general transformation equations to convert Cartesian coordinates to generalized coordinates are

$$\begin{aligned} q_1 &= q_1(x_1, x_2, \ldots, x_n, t) \\ q_2 &= q_2(x_1, x_2, \ldots, x_n, t) \\ &\vdots \\ q_n &= q_n(x_1, x_2, \ldots, x_n, t) \end{aligned} \tag{1}$$

The inverse transformation equations to convert generalized coordinates to Cartesian coordinates are

$$\begin{aligned} x_1 &= x_1(q_1, q_1, \ldots, q_n, t) \\ x_2 &= y_1(q_1, q_1, \ldots, q_n, t) \\ &\vdots \\ x_n &= z_N(q_1, q_1, \ldots, q_n, t) \end{aligned} \tag{2}$$

The inverse transformation given in equation (2) is possible only if the Jacobian determinant of equation (1) is non-zero, i.e.

$$\frac{\partial(q_1, q_2, \ldots, q_n)}{\partial(x_1, x_2, \ldots, x_n)} = \begin{vmatrix} \frac{\partial q_1}{\partial x_1} & \frac{\partial q_1}{\partial x_2} & \cdots & \frac{\partial q_1}{\partial x_n} \\ \frac{\partial q_2}{\partial x_1} & \frac{\partial q_2}{\partial x_2} & \cdots & \frac{\partial q_2}{\partial x_n} \\ & & \vdots & \\ \frac{\partial q_n}{\partial x_1} & \frac{\partial q_n}{\partial x_2} & \cdots & \frac{\partial q_n}{\partial x_n} \end{vmatrix} \neq 0$$

The generalized coordinates are usually assumed to be *linearly independent*, and thus form a *minimal set* of coordinates to describe a problem.

2.2 Generalized velocities

We know that any arbitrary Cartesian coordinate x_i is related to the generalized coordinates q_j as

$$x_i = x_i(q_1, q_2, \ldots, q_n, t)$$

The component of the velocity of a particle in the x_i direction is

$$v_i = \frac{\mathrm{d}x_i}{\mathrm{d}t}$$

By the chain rule, we have

$$\begin{aligned} v_i &= \frac{\partial x_i}{\partial q_1}\frac{\mathrm{d}q_1}{\mathrm{d}t} + \frac{\partial x_i}{\partial q_2}\frac{\mathrm{d}q_2}{\mathrm{d}t} + \cdots + \frac{\partial x_i}{\partial q_n}\frac{\mathrm{d}q_n}{\mathrm{d}t} + \frac{\partial x_1}{\partial t} \\ &= \sum_{k=1}^{n} \frac{\partial x_i}{\partial q_k}\frac{\mathrm{d}q_k}{\mathrm{d}t} + \frac{\partial x_1}{\partial t} \\ \implies v_i &= \sum_{k=1}^{n} \frac{\partial x_i}{\partial q_k}\dot{q}_k + \frac{\partial x_1}{\partial t} \end{aligned} \tag{3}$$

In equation (3), the quantities $\dot{q}_k = \frac{\mathrm{d}q_k}{\mathrm{d}t}$ are known as *generalized velocities*.

Now, since a mixed second-order partial derivative does not depend on the order in which the derivatives are taken, we can write

$$\frac{\partial}{\partial \dot{q}_j}\frac{\partial x_i}{\partial t} = \frac{\partial}{\partial t}\frac{\partial x_i}{\partial \dot{q}_j}$$

Because x_i is independent of the generalized velocities $\dot{q}_k$, we have $\frac{\partial x_i}{\partial \dot{q}_j} = 0$. Hence, we have

$$\frac{\partial}{\partial \dot{q}_j}\frac{\partial x_i}{\partial t} = 0 \tag{4}$$

Differentiating equation (3) with respect to $\dot{q}_j$, we obtain

$$\begin{aligned}
\frac{\partial v_i}{\partial \dot{q}_j} &= \frac{\partial}{\partial \dot{q}_j}\sum_{k=1}^{n}\frac{\partial x_i}{\partial q_k}\dot{q}_k + \frac{\partial}{\partial \dot{q}_j}\frac{\partial x_1}{\partial t} \\
&= \frac{\partial}{\partial \dot{q}_j}\sum_{k=1}^{n}\frac{\partial x_i}{\partial q_k}\dot{q}_k + 0 \quad \text{[using equation (4)]} \\
&= \sum_{k=1}^{n}\left(\frac{\partial}{\partial \dot{q}_j}\frac{\partial x_i}{\partial q_k}\right)\dot{q}_k + \sum_{k=1}^{n}\frac{\partial x_i}{\partial q_k}\frac{\partial \dot{q}_k}{\partial \dot{q}_j} \\
&= 0 + \sum_{k=1}^{n}\frac{\partial x_i}{\partial q_k}\delta_{jk} \\
\implies \frac{\partial v_i}{\partial \dot{q}_j} &= \frac{\partial x_i}{\partial q_j} \qquad (5)
\end{aligned}$$

Equation (5) implies that *the Cartesian velocity is related to the generalized velocity in the same way as the Cartesian coordinate is related to the generalized coordinate.*

Note: The dimensions of the generalized velocities need not be those of length per time squared, and may even be dimensionless.

2.3 Degrees of freedom

The number of degrees of freedom is the number of independent coordinates needed to completely specify the position of every part of the system.

- To describe the position of a single free particle ($N = 1$) one must specify the values of three coordinates (say x, y and z). Thus, for a free particle, the degrees of freedom is $f = 3$.
- For a system of two free particles ($N = 2$), we need to specify the positions of both particles. Each particle has three degrees of freedom, so for the system as a whole, the degrees of freedom is $f = 6$.
- In general, for a mechanical system consisting of N free particles, the degrees of freedom is $f = 3N$, which is the maximum possible degrees of freedom.

For a given system, the number of generalized coordinates always equals its degrees of freedom, i.e. $n = f$.

2.4 Constraints

Constraints on a system of particles are physical limitations or mathematical conditions restricting the positions or velocities of particles, thereby reducing their total degrees of freedom.

When constraints act on a system, its particles are no longer free. Therefore the degrees of freedom reduce by the number of constraints acting on the system.

Constraints are relationships between coordinates of a system. We can illustrate this using the following examples.

a) Consider a marble that is free to move on the surface of a table which is at a height of 1 m from the ground. Its motion can be completely described using the x and y coordinates, because it is constrained to move on the plane surface of the table. The constraint on the marble can then be expressed as $z = 1$.

   So, instead of having $f = 3$ degrees of freedom, the marble now has $f = 2$ degrees of freedom.

b) Consider a bead on a stretched string. Its motion can be completely described using the x coordinate alone, because it is constrained to move along the length of the string. The constraints on the bead can then be expressed as $y = 0$ and $z = 0$.

   So, instead of having $f = 3$ degrees of freedom, the bead now has $f = 1$ degree of freedom.

In both cases above, we find that the degrees of freedom reduces by the number of constraints. In the case of the marble on the table, there is one constraint acting on it, and thereby the degrees of freedom becomes $f = 3-1 = 2$. Likewise, in the case of the bead on the string, due to two constraints acting on it, the degrees of freedom becomes $f = 3-2 = 1$.

In general, if a system of N particles is acted upon by k constraints, then the degrees of freedom of the system becomes $f = 3N - k$. Hence, the number of generalized coordinates required to describe the motion is $n = f = 3N - k$.

Therefore,

- Number of particles: N
- Number of Cartesian coordinates: $3N$
- Number of constraints: k
- Number of generalized coordinates: $n = 3N - k$

2.4.1 Types of constraints

- **Holonomic versus non-holonomic**: This classification relates to how the constraint depends on the coordinates and velocities.

  1. *Holonomic constraints*: A constraint is holonomic if it can be written as an equation involving the coordinates of the system (and possibly time), but not their velocities. They have the general form:

     $$f(q_1, q_2, \ldots, q_n, t) = 0$$

     Holonomic constraints can be expressed purely in terms of coordinates.

  2. *Non-holonomic constraints*: A constraint is non-holonomic if it cannot be expressed solely as a coordinate equation. Often these involve *velocities* or are *inequalities* that are not integrable into a coordinate relationship.

     In non-holonomic constraints, the allowed motion depends on the path the system takes, not just its instantaneous position.

- **Scleronomous versus rheonomous**: This classification relates to whether the constraint depends on time explicitly.

  1. *Scleronomous constraints*: A constraint is scleronomic if it does not depend explicitly on time, i.e. the equation describing it contains only the coordinates (and possibly velocities), not time itself. These describe fixed constraints.
     For example, a simple pendulum of fixed length has the constraint $\sqrt{x^2+y^2}-L=0$, which does not involve time explicitly. So this constraint is holonomic as well as schleronomous.
  2. *Rheonomous constraints*:A constraint is rheonomic if it explicitly depends on time, i.e. the constraint equation includes time as a variable. Rheonomic constraints describe moving or time-dependent limitations on motion.
     For example, a pendulum whose point of support is itself moving in time (e.g. oscillating up and down) has a constraint that depends on time, and is therefore rheonomous.

2.5 Virtual displacements

A virtual displacement δx_i is defined as an infinitesimal, instantaneous displacement of the coordinate x_i which is consistent with any constraints acting on the system.

Note: we distinguish between real and virtual displacements as follows:

- δx_i – virtual displacement
- $\mathrm{d}x_i$ – real displacement

Consider the following transformation equation for an arbitrary Cartesian coordinate:

$$x_i = x_i(q_1, q_2, \ldots, q_n, t) \tag{1}$$

Taking the differential of equation (1), we obtain

$$\begin{aligned} \mathrm{d}x_i &= \frac{\partial x_i}{\partial q_1}\mathrm{d}q_1 + \frac{\partial x_i}{\partial q_2}\mathrm{d}q_2 + \cdots + \frac{\partial x_i}{\partial q_n}\mathrm{d}q_n + \frac{\partial x_i}{\partial t}\mathrm{d}t \\ \implies \mathrm{d}x_i &= \sum_{\alpha=1}^{n}\frac{\partial x_i}{\partial q_\alpha}\mathrm{d}q_\alpha + \frac{\partial x_i}{\partial t}\mathrm{d}t \end{aligned} \tag{2}$$

However, a virtual displacement is defined to be instantaneous i.e. $\mathrm{d}t = 0$. Hence we obtain the virtual displacement using equation (2) as

$$\delta x_i = \sum_{\alpha=1}^{n}\frac{\partial x_i}{\partial q_\alpha}\delta q_\alpha \tag{3}$$

2.6 Virtual work and generalized force

Suppose a system is subjected to a number of applied forces. Let F_i be the component of the force along the coordinate axis x_i.

Then by allowing all the Cartesian coordinates to undergo virtual displacements δx_i, the virtual work performed by the applied forces will be

$$\delta W = \sum_{i=1}^{3N} F_i \delta x_i \tag{4}$$

Substituting for δx_i from equation (3) in equation (4), we obtain

$$\begin{aligned} \delta W &= \sum_{i=1}^{3N} F_i \left(\sum_{\alpha=1}^{n} \frac{\partial x_i}{\partial q_\alpha} \delta q_\alpha \right) \\ &= \sum_{\alpha=1}^{n} \left(\sum_{i=1}^{3N} F_i \frac{\partial x_i}{\partial q_\alpha} \right) \delta q_\alpha \end{aligned} \tag{5}$$

We define the *generalized force* as

$$Q_\alpha \equiv \sum_{i=1}^{3N} F_i \frac{\partial x_i}{\partial q_\alpha} \tag{6}$$

Substituting equation (6) into equation (5), we obtain the expression of virtual work in terms of generalized forces and displacements to be

$$\boxed{\delta W = \sum_{\alpha=1}^{n} Q_\alpha \delta q_\alpha} \tag{7}$$

Note: The dimensions of the generalized forces need not be those of force, and may even be dimensionless.

2.6.1 The principle of virtual work

Consider a system of N particles which is in equilibrium. Then the total force on any arbitrary particle vanishes, i.e. $\mathbf{F}_i = \mathbf{0}$.

Now, if the virtual displacement of the ith particle is $\delta \mathbf{r}_i$, then the virtual work on it due to the force $\mathbf{F}_i$ also vanishes, i.e. $\delta W_i = \mathbf{F}_i \cdot \delta \mathbf{r}_i = 0$. Hence, the sum of virtual work done on all particles must also vanish, i.e.

$$\sum_{i=1}^{N} \mathbf{F}_i \cdot \delta \mathbf{r}_i = 0 \tag{1}$$

We can now decompose the force acting on any arbitrary particle into the applied force $\mathbf{F}_i^{(a)}$ and the force due to the constraints $\mathbf{f}_i$, i.e.

$$\mathbf{F}_i = \mathbf{F}_i^{(a)} + \mathbf{f}_i \tag{2}$$

Inserting equation (2) into equation (1) gives us

$$\sum_{i=1}^{N}\left(\mathbf{F}_i^{(a)}+\mathbf{f}_i\right)\cdot\delta\mathbf{r}_i=0$$
$$\implies \sum_{i=1}^{N}\mathbf{F}_i^{(a)}\cdot\delta\mathbf{r}_i+\sum_{i=1}^{N}\mathbf{f}_i\cdot\delta\mathbf{r}_i=0 \tag{3}$$

Because we restrict ourselves only to those systems where the constraint forces do no work (such as a rigid body), we must have that

$$\sum_{i=1}^{N}\mathbf{f}_i\cdot\delta\mathbf{r}_i=0$$

Therefore, we obtain from equation (3)

$$\boxed{\sum_{i=1}^{N}\mathbf{F}_i^{(a)}\cdot\delta\mathbf{r}_i=0} \tag{4}$$

Equation (4) denotes the **principle of virtual work**, which states that *for a system in equilibrium, the virtual work done by all applied forces is zero.*

2.7 D'Alembert's principle

Consider a system of N particles which are acted upon by various applied and constraint forces. According to Newton's second law of motion, the force acting on an arbitrary i^{th} particle is given by

$$\mathbf{F}_i=\frac{\mathrm{d}\mathbf{p}_i}{\mathrm{d}t}=\dot{\mathbf{p}}_i$$
$$\implies \mathbf{F}_i-\dot{\mathbf{p}}_i=\mathbf{0},\quad i=1,2,\ldots,N$$

The above set of equations imply that any i^{th} particle in a system is in equilibrium under a force, which equals the resultant of the actual force $\mathbf{F}_i$ and an inertial force $-\dot{\mathbf{p}}_i$.

Hence for virtual displacements $\delta\mathbf{r}_i$, the net virtual work on the system also vanishes, i.e.

$$\sum_{i=1}^{N}(\mathbf{F}_i-\dot{\mathbf{p}}_i)\cdot\delta\mathbf{r}_i=0 \tag{1}$$

But the force on the ith particle is due to the applied force $\mathbf{F}_i^{(a)}$ and the force due to the constraints $\mathbf{f}_i$, i.e.

$$\mathbf{F}_i=\mathbf{F}_i^{(a)}+\mathbf{f}_i$$

This gives us, from equation (1)

$$\sum_{i=1}^{N}\left(\mathbf{F}_i^{(a)}+\mathbf{f}_i-\dot{\mathbf{p}}_i\right)\cdot\delta\mathbf{r}_i=0$$

$$\implies \sum_{i=1}^{N}\left(\mathbf{F}_i^{(a)}-\dot{\mathbf{p}}_i\right)\cdot\delta\mathbf{r}_i+\sum_{i=1}^{N}\mathbf{f}_i\cdot\delta\mathbf{r}_i=0 \tag{2}$$

Restricting ourselves to systems where the constraint forces do no work, i.e. $\sum_{i=1}^{N}\mathbf{f}_i\cdot\delta\mathbf{r}_i=0$, we have from equation (2)

$$\sum_{i=1}^{N}\left(\mathbf{F}_i^{(a)}-\dot{\mathbf{p}}_i\right)\cdot\delta\mathbf{r}_i=0$$

Because the forces due to the constraints do not appear, we can drop the superscript $^{(a)}$ in the above equation without ambiguity and re-write the same as

$$\boxed{\sum_{i=1}^{N}(\mathbf{F}_i-\dot{\mathbf{p}}_i)\cdot\delta\mathbf{r}_i=0} \tag{3}$$

Equation (3) denotes the **D'Alembert's principle**, which states that *the total virtual work done by the applied forces and the inertial forces on a system of particles is zero for any virtual displacement consistent with the constraints of the system.*

3 Lagrangian formalism

3.1 The Lagrangian

The Lagrangian $\mathscr{L}$ is a function that describes the dynamics of a physical system in terms of energy. Mathematically, it is the excess of the kinetic energy of a system over its potential energy, i.e.

$$\mathscr{L}(q_i, \dot{q}_i, t) = T(\dot{q}_i, t) - V(q_i, t), \quad i = 1, 2, \ldots, n$$
$$\text{or, } \mathscr{L} = T - V$$

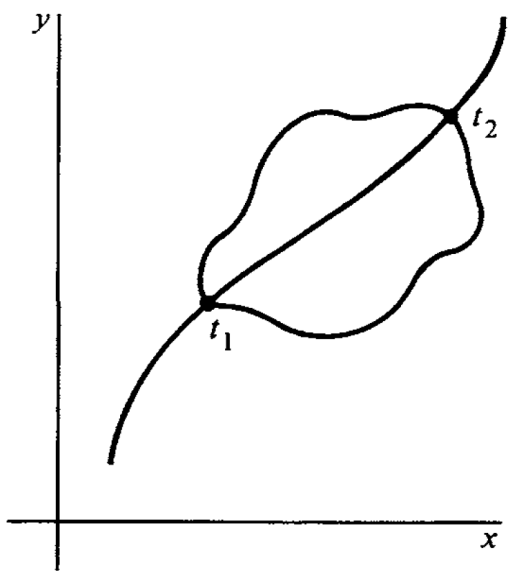
where the kinetic energy T is a function of generalized velocities and time, and the potential energy V is a function of generalized coordinates and time.

Action integral

Given a system that is described by the Lagrangian $\mathscr{L}$, its action integral I is defined as the line integral of $\mathscr{L}$ during the motion of the system from time t_1 to time t_2, i.e.

$$I = \int_{t_1}^{t_2} \mathscr{L}(q_i, \dot{q}_i, t)\,\mathrm{d}t$$

3.2 Hamilton's principle



The motion of the system from time t_1 to time t_2 is such that the action integral has a stationary value for the actual path of the motion, i.e.

$$\boxed{\delta I = \delta \int_{t_1}^{t_2} \mathscr{L}(q_i, \dot{q}_i, t)\,\mathrm{d}t = 0}$$

In other words, out of all possible paths by which the system point could travel from its position at time t_1 to its position at time t_2, it will actually travel along that path for which the value of the action integral is stationary.

3.3 Lagrange's equations

According to D'Alembert's principle, we have

$$\begin{aligned}
\sum_{i=1}^{N}(\mathbf{F}_i - \dot{\mathbf{p}}_i)\cdot\delta\mathbf{r}_i &= 0 \\
\implies \sum_{i=1}^{N}\mathbf{F}\cdot\delta\mathbf{r}_i - \sum_{i=1}^{N}\dot{\mathbf{p}}_i\cdot\delta\mathbf{r}_i &= 0 \qquad (1)
\end{aligned}$$

Let us consider the first term in equation (1). We have

$$\begin{aligned}
\sum_{i=1}^{N}\mathbf{F}\cdot\delta\mathbf{r}_i &= \sum_{i=1}^{N}\mathbf{F}\cdot\sum_{j=1}^{n}\frac{\partial\mathbf{r}_i}{\partial q_j}\delta q_j \\
&= \sum_{j=1}^{n}\underbrace{\left(\sum_{i=1}^{N}\mathbf{F}_i\cdot\frac{\partial\mathbf{r}_i}{\partial q_j}\right)}_{=Q_j}\delta q_j \quad \text{[interchanging summations]} \\
\implies \sum_{i=1}^{N}\mathbf{F}\cdot\delta\mathbf{r}_i &= \sum_{j=1}^{n}Q_j\delta q_j \qquad (2)
\end{aligned}$$

Similarly, we now consider the second term in equation (1). We now have

$$\begin{aligned}
\sum_{i=1}^{N}\dot{\mathbf{p}}_i\cdot\delta\mathbf{r}_i &= \sum_{i=1}^{N}(m_i\ddot{\mathbf{r}}_i)\cdot\left(\sum_{j=1}^{n}\frac{\partial\mathbf{r}_i}{\partial q_j}\delta q_j\right) \\
\implies \sum_{i=1}^{N}\dot{\mathbf{p}}_i\cdot\delta\mathbf{r}_i &= \sum_{j=1}^{n}\left(\sum_{i=1}^{N}m_i\ddot{\mathbf{r}}_i\cdot\frac{\partial\mathbf{r}_i}{\partial q_j}\right)\delta q_j \qquad (3)
\end{aligned}$$

Using the product rule for differentiation, we notice that

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\left(m_i\dot{\mathbf{r}}_i\cdot\frac{\partial\mathbf{r}_i}{\partial q_j}\right) &= m_i\ddot{\mathbf{r}}_i\cdot\frac{\partial\mathbf{r}_i}{\partial q_j} + m_i\dot{\mathbf{r}}_i\cdot\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathbf{r}_i}{\partial q_j} \\
&= m_i\ddot{\mathbf{r}}_i\cdot\frac{\partial\mathbf{r}_i}{\partial q_j} + m_i\dot{\mathbf{r}}_i\cdot\frac{\partial}{\partial q_j}\frac{\mathrm{d}\mathbf{r}_i}{\mathrm{d}t} \\
&= m_i\ddot{\mathbf{r}}_i\cdot\frac{\partial\mathbf{r}_i}{\partial q_j} + m_i\dot{\mathbf{r}}_i\cdot\frac{\partial\dot{\mathbf{r}}_i}{\partial q_j} \\
\implies m_i\ddot{\mathbf{r}}_i\cdot\frac{\partial\mathbf{r}_i}{\partial q_j} &= \frac{\mathrm{d}}{\mathrm{d}t}\left(m_i\mathbf{v}_i\cdot\frac{\partial\mathbf{r}_i}{\partial q_j}\right) - m_i\mathbf{v}_i\cdot\frac{\partial\mathbf{v}_i}{\partial q_j} \\
\implies m_i\ddot{\mathbf{r}}_i\cdot\frac{\partial\mathbf{r}_i}{\partial q_j} &= \frac{\mathrm{d}}{\mathrm{d}t}\left(m_i\mathbf{v}_i\cdot\frac{\partial\mathbf{v}_i}{\partial\dot{q}_j}\right) - m_i\mathbf{v}_i\cdot\frac{\partial\mathbf{v}_i}{\partial q_j} \qquad (4)
\end{aligned}$$

Again, we notice that

$$\begin{aligned}\frac{\partial}{\partial q_j}\left(\frac{1}{2}m_i v_i^2\right) &= \frac{\partial}{\partial q_j}\left(\frac{1}{2}m_i \mathbf{v}_i\cdot\mathbf{v}_i\right)\\ &= \frac{1}{2}m_i\times 2\mathbf{v}_i\cdot\frac{\partial \mathbf{v}_i}{\partial q_j}\\ &= m_i\mathbf{v}_i\cdot\frac{\partial \mathbf{v}_i}{\partial q_j}\end{aligned}$$

$$\begin{aligned}\frac{\partial}{\partial \dot{q}_j}\left(\frac{1}{2}m_i v_i^2\right) &= \frac{\partial}{\partial \dot{q}_j}\left(\frac{1}{2}m_i \mathbf{v}_i\cdot\mathbf{v}_i\right)\\ &= \frac{1}{2}m_i\times 2\mathbf{v}_i\cdot\frac{\partial \mathbf{v}_i}{\partial \dot{q}_j}\\ &= m_i\mathbf{v}_i\cdot\frac{\partial \mathbf{v}_i}{\partial \dot{q}_j}\end{aligned}$$

Inserting the above into equation (4), we obtain

$$\begin{aligned}m_i\ddot{\mathbf{r}}_i\cdot\frac{\partial \mathbf{r}_i}{\partial q_j} &= \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{q}_j}\left(\frac{1}{2}m_i v_i^2\right)-\frac{\partial}{\partial q_j}\left(\frac{1}{2}m_i v_i^2\right)\\ \implies \sum_{i=1}^{N} m_i\ddot{\mathbf{r}}_i\cdot\frac{\partial \mathbf{r}_i}{\partial q_j} &= \sum_{i=1}^{N}\left[\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{q}_j}\left(\frac{1}{2}m_i v_i^2\right)-\frac{\partial}{\partial q_j}\left(\frac{1}{2}m_i v_i^2\right)\right]\\ &= \sum_{i=1}^{N}\left[\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T_i}{\partial \dot{q}_j}-\frac{\partial T_i}{\partial q_j}\right]\\ &= \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{q}_j}\sum_{i=1}^{N}T_i-\frac{\partial}{\partial q_j}\sum_{i=1}^{N}T_i \quad [T_i \text{ is K.E. of } i^{\text{th}} \text{ particle}]\\ \implies \sum_{i=1}^{N} m_i\ddot{\mathbf{r}}_i\cdot\frac{\partial \mathbf{r}_i}{\partial q_j} &= \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_j}-\frac{\partial T}{\partial q_j} \quad [T \text{ is total energy of the system}]\end{aligned} \tag{5}$$

Inserting equation (5) into equation (3) now gives us

$$\sum_{i=1}^{N}\dot{\mathbf{p}}_i\cdot\delta\mathbf{r}_i = \sum_{j=1}^{n}\left[\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_j}-\frac{\partial T}{\partial q_j}\right]\delta q_j \tag{6}$$

Finally, inserting equation (2) and (6) into equation (eqn:lag-eqn:dap-terms), we have

$$\begin{aligned}\sum_{j=1}^{n}Q_j\delta q_j-\sum_{j=1}^{n}\left[\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_j}-\frac{\partial T}{\partial q_j}\right]\delta q_j &= 0\\ \implies \sum_{j=1}^{n}\left[\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_j}-\frac{\partial T}{\partial q_j}-Q_j\right]\delta q_j &= 0\end{aligned}$$

Because, δq_j is arbitrary, we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_j}-\frac{\partial T}{\partial q_j}-Q_j = 0 \tag{7}$$

Now, if we consider the system to be conservative, we would have $\mathbf{F}_i = -\nabla_i V$ for potential $V = V(q_j)$. Then we have

$$Q_j = \sum_{i=1}^{N}\mathbf{F}_i\cdot\frac{\partial \mathbf{r}_i}{\partial q_j} = -\sum_{i=1}^{N}\nabla_i V\cdot\frac{\partial \mathbf{r}_i}{\partial q_j} = -\frac{\partial V}{\partial q_j}$$

This gives us, in equation (7)

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_j} - \frac{\partial T}{\partial q_j} + \frac{\partial V}{\partial q_j} = 0 \tag{8}$$

Because the potential is assumed to be independent of the generalized velocities, we have that $\frac{\partial V}{\partial \dot{q}_j} = 0$ for all j. So we can write

$$\begin{aligned} \frac{\partial T}{\partial \dot{q}_j} = \frac{\partial T}{\partial \dot{q}_j} - \frac{\partial V}{\partial \dot{q}_j} &= \frac{\partial (T-V)}{\partial \dot{q}_j} \\ \implies \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_j} &= \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial (T-V)}{\partial \dot{q}_j} \end{aligned}$$

Therefore, equation (8) becomes

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial (T-V)}{\partial \dot{q}_j} - \frac{\partial (T-V)}{\partial q_j} &= 0 \\ \implies \boxed{\frac{\partial \mathscr{L}}{\partial \dot{q}_j} - \frac{\partial \mathscr{L}}{\partial q_j} = 0} \end{aligned} \tag{9}$$

where $\mathscr{L} = T - V$ is the Lagrangian of the system.

The set of equations given in (9) for each j is known as the Lagrange's equations for a system.

4 Physical problems and applications

1. Obtain the Lagrange's equations of motion of a particle with mass m in a potential $V(r,\theta,\phi)$ in spherical polar coordinates.
Solution: In spherical polar coordinates, the elementary lengths are $\mathrm{d}r$, $r\,\mathrm{d}\theta$ and $r\sin\theta\,\mathrm{d}\phi$.

The corresponding velocities are

$$v_r = \frac{\mathrm{d}r}{\mathrm{d}t} = \dot{r}, \quad v_\theta = r\frac{\mathrm{d}\theta}{\mathrm{d}t} = r\dot{\theta}, \quad v_\phi = r\sin\theta\frac{\mathrm{d}\phi}{\mathrm{d}t} = r\sin\theta\,\dot{\phi}$$

This gives us the speed of the particle in spherical polar coordinates to be

$$\begin{aligned} v^2 &= v_r^2 + v_\theta^2 + v_\phi^2 \\ \implies v^2 &= \dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2 \end{aligned}$$

Hence, the kinetic energy of the particle is

$$T = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2\right)$$

Assuming that the potential energy is independent of velocities, we can write $V = V(r,\theta,\phi)$. This gives us the Lagrangian to be

$$\begin{aligned} \mathscr{L} &= T - V \\ \implies \mathscr{L} &= \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2\right) - V(r,\theta,\phi) \end{aligned} \tag{1}$$

We choose the generalized coordinates to be the spherical coordinates, i.e. $q_1 = r$, $q_2 = \theta$ and $q_3 = \phi$. The Lagrange's equations of motion are

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathscr{L}}{\partial\dot{q}_j} - \frac{\partial\mathscr{L}}{\partial q_j} = 0, \quad j = 1,2,3 \tag{2}$$

We then obtain the equations of motion in spherical polar coordinates as follows:

- r-coordinate:
The Lagrange's equation (2) for $q_1 = r$ is

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathscr{L}}{\partial\dot{q}_1} - \frac{\partial\mathscr{L}}{\partial q_1} &= 0 \\ \implies \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathscr{L}}{\partial\dot{r}} - \frac{\partial\mathscr{L}}{\partial r} &= 0 \end{aligned} \tag{3}$$

From equation (1), we have

$$\begin{aligned} \frac{\partial\mathscr{L}}{\partial\dot{r}} &= 2\times\frac{1}{2}m\dot{r} = m\dot{r} \\ \implies \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathscr{L}}{\partial\dot{r}} &= \frac{\mathrm{d}}{\mathrm{d}t}(m\dot{r}) = m\ddot{r} \end{aligned}$$

and also

$$\begin{aligned} \frac{\partial\mathscr{L}}{\partial r} &= 2\times\frac{1}{2}mr\dot{\theta}^2 + 2\times\frac{1}{2}mr\sin^2\theta\,\dot{\phi}^2 - \frac{\partial V}{\partial r} \\ &= mr\dot{\theta}^2 + mr\sin^2\theta\,\dot{\phi}^2 - \frac{\partial V}{\partial r} \end{aligned}$$

Therefore, in equation (3), we obtain

$$m\ddot{r} - mr\dot{\theta}^2 - mr\sin^2\theta\dot{\phi}^2 + \frac{\partial V}{\partial r} = 0$$
$$\Longrightarrow \boxed{m\ddot{r} - mr\left(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2\right) + \frac{\partial V}{\partial r} = 0}$$

- θ-coordinate:
  The Lagrange's equation (2) for $q_2 = \theta$ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{q}_2} - \frac{\partial \mathscr{L}}{\partial q_2} = 0$$
$$\Longrightarrow \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{\theta}} - \frac{\partial \mathscr{L}}{\partial \theta} = 0 \tag{4}$$

  From equation (1), we have

$$\frac{\partial \mathscr{L}}{\partial \dot{\theta}} = 2 \times \frac{1}{2} m r^2 \dot{\theta} = m r^2 \dot{\theta}$$
$$\Longrightarrow \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{\theta}} = \frac{\mathrm{d}}{\mathrm{d}t}\left(m r^2 \dot{\theta}\right) = m r^2 \ddot{\theta} + 2 m r \dot{r} \dot{\theta}$$

  and also

$$\frac{\partial \mathscr{L}}{\partial \theta} = 2 \times \frac{1}{2} m r^2 \sin\theta\cos\theta\dot{\phi}^2 - \frac{\partial V}{\partial \theta}$$
$$= m r^2 \sin\theta\cos\theta\dot{\phi}^2 - \frac{\partial V}{\partial \theta}$$

  Therefore, in equation (4), we obtain

$$\boxed{m r^2 \ddot{\theta} + 2 m r \dot{r} \dot{\theta} - m r^2 \sin\theta\cos\theta\dot{\phi}^2 + \frac{\partial V}{\partial \theta} = 0}$$

- ϕ-coordinate:
  The Lagrange's equation (2) for $q_3 = \phi$ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{q}_3} - \frac{\partial \mathscr{L}}{\partial q_3} = 0$$
$$\Longrightarrow \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{\phi}} - \frac{\partial \mathscr{L}}{\partial \phi} = 0 \tag{5}$$

  From equation (1), we have

$$\frac{\partial \mathscr{L}}{\partial \dot{\phi}} = 2 \times \frac{1}{2} m r^2 \sin^2\theta\dot{\phi} = m r^2 \sin^2\theta\dot{\phi}$$
$$\Longrightarrow \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{\phi}} = \frac{\mathrm{d}}{\mathrm{d}t}\left(m r^2 \sin^2\theta\dot{\phi}\right) = m r^2 \sin^2\theta\ddot{\phi} + 2 m r^2 \sin\theta\cos\theta\dot{\theta}\dot{\phi} + 2 m r \sin^2\theta\dot{r}\dot{\phi}$$

  and also

$$\frac{\partial \mathscr{L}}{\partial \phi} = -\frac{\partial V}{\partial \phi}$$

  Therefore, in equation (5), we obtain

$$\boxed{m r^2 \sin^2\theta\ddot{\phi} + 2 m r^2 \sin\theta\cos\theta\dot{\theta}\dot{\phi} + 2 m r \sin^2\theta\dot{r}\dot{\phi} + \frac{\partial V}{\partial \phi} = 0}$$

2. Obtain the Lagrange's equations of motion of a particle with mass m in a potential $V(\rho,\phi,z)$ in cylindrical polar coordinates.
Solution: In cylindrical polar coordinates, the elementary lengths are $\mathrm{d}\rho$, $\rho\,\mathrm{d}\phi$ and $\mathrm{d}z$.

The corresponding velocities are

$$v_\rho = \frac{\mathrm{d}\rho}{\mathrm{d}t} = \dot{\rho}, \quad v_\phi = \rho\frac{\mathrm{d}\phi}{\mathrm{d}t} = \rho\dot{\phi}, \quad v_z = \frac{\mathrm{d}z}{\mathrm{d}t} = \dot{z}$$

This gives us the speed of the particle in cylindrical polar coordinates to be

$$\begin{aligned} v^2 &= v_\rho^2 + v_\phi^2 + v_z^2 \\ \implies v^2 &= \dot{\rho}^2 + \rho^2\dot{\phi}^2 + \dot{z}^2 \end{aligned}$$

Hence, the kinetic energy of the particle is

$$T = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\dot{\rho}^2 + \rho^2\dot{\phi}^2 + \dot{z}^2\right)$$

Assuming that the potential energy is independent of velocities, we can write $V = V(\rho,\phi,z)$. This gives us the Lagrangian to be

$$\begin{aligned} \mathscr{L} &= T - V \\ \implies \mathscr{L} &= \frac{1}{2}m\left(\dot{\rho}^2 + \rho^2\dot{\phi}^2 + \dot{z}^2\right) - V(\rho,\phi,z) \end{aligned} \tag{1}$$

We choose the generalized coordinates to be the cylindrical coordinates, i.e. $q_1 = \rho$, $q_2 = \phi$ and $q_3 = z$. The Lagrange's equations of motion are

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathscr{L}}{\partial\dot{q}_j} - \frac{\partial\mathscr{L}}{\partial q_j} = 0, \quad j = 1,2,3 \tag{2}$$

We then obtain the equations of motion in cylindrical polar coordinates as follows:

- ρ-coordinate:
The Lagrange's equation (2) for $q_1 = \rho$ is

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathscr{L}}{\partial\dot{q}_1} - \frac{\partial\mathscr{L}}{\partial q_1} &= 0 \\ \implies \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathscr{L}}{\partial\dot{\rho}} - \frac{\partial\mathscr{L}}{\partial\rho} &= 0 \end{aligned} \tag{3}$$

From equation (1), we have

$$\begin{aligned} \frac{\partial\mathscr{L}}{\partial\dot{\rho}} &= 2\times\frac{1}{2}m\dot{\rho} = m\dot{\rho} \\ \implies \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathscr{L}}{\partial\dot{\rho}} &= \frac{\mathrm{d}}{\mathrm{d}t}\left(m\dot{\rho}\right) = m\ddot{\rho} \end{aligned}$$

and also

$$\begin{aligned} \frac{\partial\mathscr{L}}{\partial\rho} &= 2\times\frac{1}{2}m\rho\dot{\phi}^2 - \frac{\partial V}{\partial\rho} \\ &= m\rho\dot{\phi}^2 - \frac{\partial V}{\partial\rho} \end{aligned}$$

Therefore, in equation (3), we obtain

$$\boxed{m\ddot{\rho} - m\rho\dot{\phi}^2 + \frac{\partial V}{\partial\rho} = 0}$$

- ϕ-coordinate:
  The Lagrange's equation (2) for $q_2 = \phi$ is

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{q}_2} - \frac{\partial \mathscr{L}}{\partial q_2} &= 0 \\ \implies \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{\phi}} - \frac{\partial \mathscr{L}}{\partial \phi} &= 0 \end{aligned} \tag{4}$$

  From equation (1), we have

$$\begin{aligned} \frac{\partial \mathscr{L}}{\partial \dot{\phi}} &= 2 \times \frac{1}{2} m\rho^2\dot{\phi} = m\rho^2\dot{\phi} \\ \implies \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{\phi}} &= \frac{\mathrm{d}}{\mathrm{d}t}\left(m\rho^2\dot{\phi}\right) = m\rho^2\ddot{\phi} + 2m\rho\dot{\rho}\dot{\phi} \end{aligned}$$

  and also

$$\frac{\partial \mathscr{L}}{\partial \phi} = -\frac{\partial V}{\partial \phi}$$

  Therefore, in equation (4), we obtain

$$\boxed{m\rho^2\ddot{\phi} + 2m\rho\dot{\rho}\dot{\phi} + \frac{\partial V}{\partial \phi} = 0}$$

- z-coordinate:
  The Lagrange's equation (2) for $q_3 = z$ is

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{q}_3} - \frac{\partial \mathscr{L}}{\partial q_3} &= 0 \\ \implies \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{z}} - \frac{\partial \mathscr{L}}{\partial z} &= 0 \end{aligned} \tag{5}$$

  From equation (1), we have

$$\begin{aligned} \frac{\partial \mathscr{L}}{\partial \dot{z}} &= 2 \times \frac{1}{2} m\dot{z} = m\dot{z} \\ \implies \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{z}} &= m\ddot{z} \end{aligned}$$

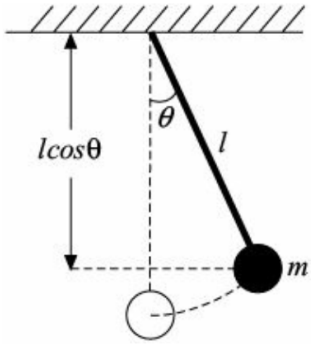
  and also

$$\frac{\partial \mathscr{L}}{\partial z} = -\frac{\partial V}{\partial z}$$

  Therefore, in equation (5), we obtain

$$\boxed{m\ddot{z} + \frac{\partial V}{\partial z} = 0}$$

3. Find the Lagrange's equations of motion of the bob of a simple pendulum.
Solution:



Consider a pendulum with a bob of mass m, suspended from a fixed ceiling with a massless string of length l. Here the number of particles in the system is $N = 1$.

The motion of the bob has the following constraints:

(a) The bob is free to perform oscillations in a plane only, say the xy-plane, and hence

$$z = 0$$

(b) The bob is always at a distance l from the point of suspension, and hence

$$x^2 + y^2 = l^2$$

Hence, the number of constraints is $k = 2$, which gives the degrees of freedom to be $n = 3N - k = 3 \times 1 - 2 = 1$. Therefore, the required number of generalized coordinates is also 1.

We can choose the generalized coordinate $q_1 = \theta$, where θ is the angle made by the pendulum string with the vertical axis.

Then the generalized velocity $\dot{\theta}$ would be the angular velocity of the bob about the point of suspension. Hence, the speed of the bob at any instant is

$$v = l\dot{\theta}$$

This gives the kinetic energy of the bob to be

$$T = \frac{1}{2}mv^2 = \frac{1}{2}ml^2\dot{\theta}^2$$

The height of the bob above the lowest point is

$$h = l - l\cos\theta = l(1 - \cos\theta)$$

So the potential energy of the bob due to gravity is

$$V = mgh = mgl(1 - \cos\theta)$$

We obtain the Lagrangian of the system as

$$\mathscr{L} = T - V$$

$$\implies \mathscr{L} = \frac{1}{2}ml^2\dot{\theta}^2 - mgl(1 - \cos\theta) \tag{1}$$

Then, the Lagrange's equation for the simple pendulum is

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{q}_1} - \frac{\partial \mathscr{L}}{\partial q_1} &= 0 \\ \implies \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{\theta}} - \frac{\partial \mathscr{L}}{\partial \theta} &= 0 \end{aligned} \tag{2}$$

From equation (1), we obtain

$$\begin{aligned} \frac{\partial \mathscr{L}}{\partial \dot{\theta}} &= 2 \times \frac{1}{2} m l^2 \dot{\theta} = m l^2 \dot{\theta} \\ \implies \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{\theta}} &= m l^2 \ddot{\theta} \end{aligned}$$

$$\begin{aligned} \frac{\partial \mathscr{L}}{\partial \theta} &= \frac{\partial}{\partial \theta}\left[-mgl(1-\cos\theta)\right] \\ &= \frac{\partial}{\partial \theta}\left[-mgl + mgl\cos\theta\right] \\ &= -mgl\sin\theta \end{aligned}$$

Substituting in equation (2), we obtain

$$\begin{aligned} m l^2 \ddot{\theta} + mgl\sin\theta &= 0 \\ \implies \ddot{\theta} + \left(\frac{g}{l}\right)\sin\theta &= 0 \end{aligned} \tag{3}$$

Equation (3) is the equation of motion for the bob of the simple pendulum.

For small oscillations, we can consider θ to be small and so we can use the small-angle approximation, i.e.

$$\sin\theta \approx \theta$$

Then equation (3) becomes as follows:

$$\begin{aligned} \ddot{\theta} + \left(\frac{g}{l}\right)\theta &= 0 \\ \implies \frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} &= -\left(\frac{g}{l}\right)\theta \end{aligned} \tag{4}$$

Comparing equation (4) with the ODE for simple harmonic motion (SHM), i.e.

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} = -\omega^2 y$$

we find that the pendulum bob also performs SHM with an angular frequency

$$\begin{aligned} \omega^2 &= \frac{g}{l} \\ \implies \omega &= \sqrt{\frac{g}{l}} \end{aligned}$$

The time period of the simple pendulum is

$$\begin{aligned} T &= \frac{2\pi}{\omega} \\ \implies T &= 2\pi\sqrt{\frac{l}{g}} \end{aligned}$$

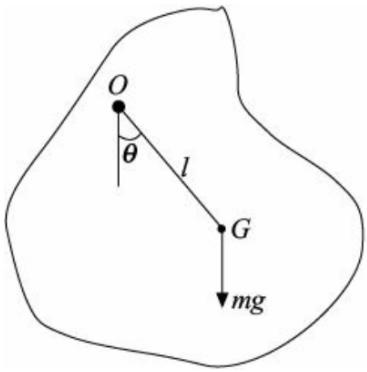
4. A rigid body capable of oscillating in a vertical plane about a fixed horizontal axis is called a compound pendulum.

   (i) Set up its Lagrangian.

   (ii) Obtain its equations of motion.

   (iii) Find the time period of the pendulum.

   Solution: Let us consider a rigid body that is fixed at point O, through which the axis of



oscillations passes. Let m be its mass, G its centre of mass which is at a distance l from O and I its moment of inertia about the axis of oscillation.

Because the body is rigid, the separation between any two arbitrary points in the body always remains constant. Hence, knowing the coordinates of any single point is sufficient to completely describe the motion of the entire body. So we consider the particle located at the centre of gravity G to be representative of the entire body.

Therefore, similar to the case of the simple pendulum, the degrees of freedom is 1, and we can choose the generalized coordinate $q_1 = \theta$, where θ is the angle made by the line OG with the vertical axis.

The rotational kinetic energy of the body is

$$T = \frac{1}{2} I \dot{\theta}^2$$

With respect to the point of oscillation, the potential energy of the rigid body is

$$V = -mgl\cos\theta$$

Therefore, the Lagrangian of the compound pendulum is

$$\begin{aligned} \mathscr{L} &= T - V \\ \implies \mathscr{L} &= \frac{1}{2} I \dot{\theta}^2 + mgl\cos\theta \end{aligned} \tag{1}$$

Then, the Lagrange's equation for the compound pendulum is

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{q}_1} - \frac{\partial \mathscr{L}}{\partial q_1} &= 0 \\ \implies \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{\theta}} - \frac{\partial \mathscr{L}}{\partial \theta} &= 0 \end{aligned} \tag{2}$$

From equation (1), we obtain

$$\frac{\partial \mathscr{L}}{\partial \dot{\theta}} = 2 \times \frac{1}{2} I\dot{\theta} = I\dot{\theta}$$
$$\implies \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathscr{L}}{\partial \dot{\theta}} = I\ddot{\theta}$$

$$\frac{\partial \mathscr{L}}{\partial \theta} = \frac{\partial}{\partial \theta}\left[mgl\cos\theta\right]$$
$$= -mgl\sin\theta$$

Substituting in equation (2), we obtain

$$I\ddot{\theta} + mgl\sin\theta = 0$$
$$\implies \ddot{\theta} + \left(\frac{mgl}{I}\right)\sin\theta = 0 \tag{3}$$

Equation (3) is the equation of motion for the compound pendulum.

Using the small-angle oscillations, we have $\sin\theta \approx \theta$. Then equation (3) becomes as follows:

$$\ddot{\theta} = -\left(\frac{mgl}{I}\right)\theta$$
$$\implies \frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = -\left(\frac{mgl}{I}\right)\theta \tag{4}$$

Comparing equation (4) with the ODE for simple harmonic motion (SHM), i.e.

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} = -\omega^2 y$$

we find that the compound pendulum also performs SHM with an angular frequency

$$\omega^2 = \frac{mgl}{I}$$
$$\implies \omega = \sqrt{\frac{mgl}{I}}$$

The time period of the compound pendulum is

$$T = \frac{2\pi}{\omega}$$
$$\implies T = 2\pi\sqrt{\frac{I}{mgl}}$$

If k is the radius of gyration of the compound pendulum, then we have $I = m(k^2+l^2)$. Then the time period of the compound pendulum becomes

$$T = 2\pi\sqrt{\frac{m(k^2+l^2)}{mgl}}$$
$$\implies T = 2\pi\sqrt{\frac{k^2+l^2}{gl}}$$